

SECTION 12

ELECTRICAL SPECIFICATIONS

This section contains electrical specifications and associated timing information for the MC68881/MC68882 (FPCP).

12.1 MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Supply Voltage	V_{CC}	-0.3 to +7.0	V
Input Voltage	V_{in}	-0.5 to +7.0	V
Operating Temperature	T_A	0 to 70	°C
Storage Temperature	T_{stg}	-55 to +150	°C

This device contains circuitry to protect the inputs against damage due to high static voltages or electric fields; however, it is advised that normal precautions be taken to avoid application of any voltage higher than maximum-rated voltages to this high-impedance circuit. Reliability of operation is enhanced if unused inputs are tied to an appropriate logic voltage level (e.g., either GND or V_{CC}).

12.2 THERMAL CHARACTERISTICS — PGA PACKAGE

Characteristic	Symbol	Value	Rating
Thermal Resistance — Ceramic Junction to Ambient	θ_{JA}	30*	°C/W
Junction to Case	θ_{JC}	15*	

*Estimated

12.3 POWER CONSIDERATIONS

The average chip-junction temperature, T_J , in °C can be obtained from:

$$T_J = T_A + (P_D \cdot \theta_{JA}) \quad (1)$$

where:

T_A = Ambient Temperature, °C

θ_{JA} = Package Thermal Resistance, Junction-to-Ambient, °C/W

P_D = $P_{INT} + P_{I/O}$

P_{INT} = $I_{CC} \times V_{CC}$, Watts — Chip Internal Power

$P_{I/O}$ = Power Dissipation on Input and Output Pins — User Determined

For most applications $P_{I/O} < P_{INT}$ and can be neglected.

The following is an approximate relationship between P_D and T_J (if $P_{I/O}$ is neglected):

$$P_D = K \div (T_J + 273^\circ\text{C}) \quad (2)$$

Solving equations (1) and (2) for K gives:

$$K = P_D \cdot (T_A + 273^\circ\text{C}) + \theta_{JA} \cdot P_D^2 \quad (3)$$

where K is a constant pertaining to the particular part. K can be determined from equation (3) by measuring P_D (at equilibrium) for a known T_A . Using this value of K, the values of P_D and T_J can be obtained by solving equations (1) and (2) iteratively for any value of T_A .

The total thermal resistance of a package (θ_{JA}) can be separated into two components, θ_{JC} and θ_{CA} , representing the barrier to heat flow from the semiconductor junction to the package (case) surface (θ_{JC}) and from the case to the outside ambient (θ_{CA}). These terms are related by the equation:

$$\theta_{JA} = \theta_{JC} + \theta_{CA} \quad (4)$$

θ_{JC} is device related and cannot be influenced by the user. However, θ_{CA} is user dependent and can be minimized by such thermal management techniques as heat sinks, ambient air cooling, and thermal convection. Thus, good thermal management on the part of the user can significantly reduce θ_{CA} so that θ_{JA} approximately equals θ_{JC} . Substitution of θ_{JC} for θ_{JA} in equation (1) will result in a lower semiconductor junction temperature.

Values for thermal resistance presented in this document, unless estimated, were derived using the procedure described in Freescale Reliability Report 7843, "Thermal Resistance Measurement Method for MC68XX Microcomponent Devices," and are provided for design purposes only. Thermal measurements are complex and dependent on procedure and setup. User derived values for thermal resistance may differ.

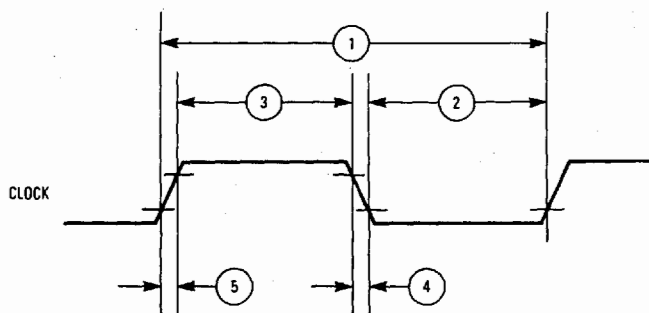
12.4 DC ELECTRICAL CHARACTERISTICS

($V_{CC} = 5.0 \text{ Vdc} \pm 5\%$; $GND = 0 \text{ Vdc}$, $T_A = 0^\circ\text{C}$ to 70°C)

Characteristic	Symbol	Min	Max	Unit
Input High Voltage	V_{IH}	2.0	V_{CC}	V
Input Low Voltage	V_{IL}	$GND - 0.5$	0.8	V
Input Leakage Current ($\approx 5.25 \text{ V}$) CLK, RESET, R/W, A0-A4, CS, DS, AS, SIZE	I_{in}	—	10	μA
Hi-Z (Off State) Input Current ($\approx 2.4 \text{ V}/0.4 \text{ V}$) DSACK0, DSACK1, D0-D31	I_{TSI}	—	20	μA
Output High Voltage ($I_{OH} = -400 \mu\text{A}$) DSACK0, DSACK1, D0-D31	V_{OH}	2.4	—	V
Output Low Voltage ($I_{OL} = 5.3 \text{ mA}$) DSACK0, DSACK1, D0-D31	V_{OL}	—	0.5	V
Output Low Current ($V_{OL} = GND$) SENSE	I_{OL}	—	500	μA
Power Dissipation	P_D	—	0.75	W
Capacitance* ($V_{IH} = 0$, $T_A = 25^\circ\text{C}$, $f = 1 \text{ MHz}$)	C_{in}	—	20	pF
Output Load Capacitance	C_L	—	130	pF

*Capacitance is periodically sampled rather than 100% tested.

Num	Characteristic	16.67 MHz		20 MHz		25 MHz		33.33 MHz		Unit
		Min	Max	Min	Max	Min	Max	Min	Max	
	Frequency of Operation	8	16.67	12.5	20	12.5	25	16.7	33.33	MHz
1	Cycle Time	60	125	50	80	40	80	30	60	ns
2,3	Clock Pulse Width (Measured from 1.5 V to 1.5 V for 33 MHz)	24	95	20	54	15	59	14	66	ns
4,5	Rise and Fall Times	—	5	—	5	—	4	—	3	ns



NOTE:

1. Timing measurements are referenced to and from a low voltage of 0.8 volts and a high voltage of 2.0 volts, unless otherwise noted. The voltage swing through this range should start outside, and pass through, the range such that the rise or fall will be linear between 0.8 volts and 2.0 volts.

Figure 12-1. Clock Input Timing Diagram

Num	Characteristic	16.67 MHz		20 MHz		25 MHz		33.33 MHz		Unit
		Min	Max	Min	Max	Min	Max	Min	Max	
6 ⁵	Address Valid to $\overline{AS}$ Asserted	15	—	10	—	5	—	5	—	ns
6A ⁵	Address Valid to $\overline{DS}$ Asserted (Read)	15	—	10	—	5	—	5	—	ns
6B ⁵	Address Valid to $\overline{DS}$ Asserted (Write)	50	—	50	—	35	—	26	—	ns
7 ⁶	$\overline{AS}$ Negated to Address Invalid	10	—	10	—	5	—	5	—	ns
7A ⁶	$\overline{DS}$ Negated to Address Invalid	10	—	10	—	5	—	5	—	ns
8 ⁹	$\overline{CS}$ Negated to $\overline{AS}$ Asserted	0	—	0	—	0	—	0	—	ns
8A ⁹	$\overline{CS}$ Negated to $\overline{DS}$ Asserted (Read)	0	—	0	—	0	—	0	—	ns
8B	$\overline{CS}$ Asserted to $\overline{DS}$ Asserted (Write)	30	—	25	—	20	—	15	—	ns
9	$\overline{AS}$ Negated to $\overline{CS}$ Negated	10	—	10	—	5	—	5	—	ns
9A	$\overline{DS}$ Negated to $\overline{CS}$ Negated	10	—	10	—	5	—	5	—	ns
10	R/W High to $\overline{AS}$ Asserted (Read)	15	—	10	—	5	—	5	—	ns
10A	R/W High to $\overline{DS}$ Asserted (Read)	15	—	10	—	5	—	5	—	ns
10B	R/W Low to $\overline{DS}$ Asserted (Write)	35	—	30	—	25	—	25	—	ns
11	$\overline{AS}$ Negated to R/W Low (Read) or $\overline{AS}$ Negated to R/W High (Write)	10	—	10	—	5	—	5	—	ns
11A	$\overline{DS}$ Negated to R/W Low (Read) or $\overline{DS}$ Negated to R/W High (Write)	10	—	10	—	5	—	5	—	ns
12	$\overline{DS}$ Width Asserted (Write)	40	—	38	—	30	—	23	—	ns
13	$\overline{DS}$ Width Negated	40	—	38	—	30	—	23	—	ns
13A ⁴	$\overline{DS}$ Negated to $\overline{AS}$ Asserted	30	—	30	—	25	—	18	—	ns
14 ²	$\overline{CS}$, $\overline{DS}$ Asserted to Data-Out Valid (Read)	—	80	—	45	—	45	—	30	ns
15	$\overline{DS}$ Negated to Data-Out Invalid (Read)	0	—	0	—	0	—	0	—	ns
16	$\overline{DS}$ Negated to Data-Out High Impedance (Read)	—	50	—	30	—	30	—	20	ns
17	Data-In Valid to $\overline{DS}$ Asserted (Write)	15	—	10	—	5	—	5	—	ns
18	$\overline{DS}$ Negated to Data-In Invalid (Write)	15	—	10	—	5	—	5	—	ns
19 ²	START True to $\overline{DSACK0}$ and $\overline{DSACK1}$ Asserted	—	50	—	35	—	25	—	20	ns
19A ⁷	$\overline{DSACK0}$ Asserted to $\overline{DSACK1}$ Asserted (Skew)	-15	15	-10	10	-10	10	—	5	ns
20	$\overline{DSACK0}$ or $\overline{DSACK1}$ Asserted to Data-Out Valid	—	50	—	43	—	32	—	17	ns
21 ⁸	START False to $\overline{DSACK0}$ and $\overline{DSACK1}$ Negated	—	50	—	30	—	30	—	20	ns
22 ⁸	START False to $\overline{DSACK0}$ and $\overline{DSACK1}$ High Impedance	—	70	—	40	—	40	—	30	ns

- Continued -

Num	Characteristic	16.67 MHz		20 MHz		25 MHz		33.33 MHz		Unit
		Min	Max	Min	Max	Min	Max	Min	Max	
23 ^{3,8}	START True to Clock High (Synchronous Read)	0	—	0	—	0	—	0	—	ns
24 ³	Clock Low to Data-Out Valid (Synchronous Read)	—	105	—	80	—	60	—	45	ns
25 ^{3,8}	START True to Data-Out Valid (Synchronous Read)	— 1.5	105+ 2.5	— 1.5	80+ 2.5	— 1.5	60+ 2.5	— 1.5	45+ 2.5	ns Clks
26 ³	Clock Low to $\overline{DSACK0}$ and $\overline{DSACK1}$ Asserted (Synchronous Read)	—	75	—	55	—	45	—	30	ns
27 ^{3,8}	START True to $\overline{DSACK0}$ and $\overline{DSACK1}$ Asserted (Synchronous Read)	— 1.5	75+ 2.5	— 1.5	55+ 2.5	— 1.5	45+ 2.5	— 1.5	30+ 2.5	ns Clks

NOTES:

- Timing measurements are referenced to and from a low voltage of 0.8 volts and a high voltage of 2.0 volts, unless otherwise noted. The voltage swing through this range should start outside, and pass through, the range such that the rise or fall will be linear between 0.8 volts and 2.0 volts.
- These specifications only apply if the MC68882 has completed all internal operations initiated by the termination of the previous bus cycle when $\overline{DS}$ was negated.
- Synchronous read cycles occur *only* when the save or response CIR locations are read.
- This specification only applies to systems in which back-to-back accesses (read-write or write-write) of the operand CIR can occur. When the MC68882 is used as a coprocessor to the MC68020/MC68030, this can occur when the addressing mode is Immediate.
- If the $\overline{SIZE}$ pin is *not* strapped to either V_{CC} or GND, it must have the same setup times as do addresses.
- If the $\overline{SIZE}$ pin is *not* strapped to either V_{CC} or GND, it must have the same hold times as do addresses.
- This number is reduced to 5 nanoseconds if $\overline{DSACK0}$ and $\overline{DSACK1}$ have equal loads.
- START is not an external signal; rather, it is the logical condition that indicates the start of an access. The logical equation for this condition is $\overline{START} = \overline{CS} + \overline{AS} + R/W \cdot \overline{DS}$.
- If a subsequent access is not a FPCP access, $\overline{CS}$ must be negated before the assertion of $\overline{AS}$ and/or $\overline{DS}$ on the non-FPCP access. These specifications replace the old specifications 8 and 8A (the old specifications implied that in all cases, transitions in $\overline{CS}$ must not occur simultaneously with transitions of $\overline{AS}$ or $\overline{DS}$. This is not a requirement of the /MC68882).

Timing diagrams (Figures 12-2, 12-3, and 12-4) are located on foldout pages at the end of this document.

